PROJECT 2

CHINOOK SQL ANALYSIS REPORT (WEEK 3)

Project Overview

This project was completed using Microsoft SOL Server and the Chinook Database to demonstrate practical SQL skills in data extraction, transformation, querying, and business analysis. The Chinook database represents a digital music store containing customer purchases, invoices, artists, albums, tracks, playlists, and media types.

The project focused on applying SQL concepts such as filtering, aggregation, joins, and window functions to solve real-world business problems and generate actionable insights.

Database Setup

The database consisted of multiple related tables including Album, Artist, Customer, Employee, Genre, Invoice, InvoiceLine, MediaType Playlist, PlaylistTrack and Track.
An Entity Relationship Diagram (ERD) was also used to visualize table relationships and understand how customer purchases connect to tracks, albums, and artists.

SQL Analysis & Insights

A. Customer Analysis

A total of 59 customers across 24 countries were analyzed. The USA recorded the highest number of customers, indicating it as the business's largest customer market.

B. Top Customers by Revenue

The highest-spending customer was Helena Holý from the Czech Republic with a total spending of $49.62, followed closely by customers from the USA, Chile, Ireland, and Hungary. This analysis helped identify high-value customers important for customer retention strategies.

C. Revenue Analysis

The total business revenue generated was $2,328.60 while the average invoice value was $5.65, indicating moderate customer spending per transaction.

Revenue by Country

The USA generated the highest revenue at $523.06, followed by:

- USA — $523.06
- Canada — $303.96
- France — $195.10
- Brazil — $190.10
- Germany — $156.48

These insights revealed the company's strongest performing markets and regions contributing most to overall sales.

D. Artist & Album Analysis

The database contained 347 albums.

Artists with Multiple Albums

The most productive artist was **Iron Maiden** with 21 albums, followed by:

• Led Zeppelin — 14 albums
• Deep Purple — 11 albums
• Metallica - 10 albums
• U2 — 10 albums

Album with the Highest Tracks

The album Greatest Hits contained the highest number of tracks with 57 songs, indicating extensive music collections within the catalog.

E. Genre & Media Analysis

Most Popular Genre

The most purchased genre was

- Rock with 835 purchases, significantly outperforming:
- Latin — 386 purchases
- Metal — 264 purchases
- Alternative & Punk - 244 purchases

This showed a strong customer preference for rock music.

Most Used Media Type

The most common media format was MPEG audio file with 3,034 tracks, demonstrating its dominance within the music store inventory.

F. Playlist & Track Analysis

Playlist with the Most Tracks:
The playlist **Music** contained the highest number of tracks with 6,580 songs, followed by 90's Music with 1,477 tracks.

Tracks Generating the Most Revenue:
The tracks Dazed and Confused and The Trooper generated the highest revenue at $4.95 each, making them the top-performing songs in the database.

SOL Techniques Applied

The project utilized several SQL concepts including:
• SELECT, WHERE, ORDER BY, TOP, YEAR, MONTH AND AS
• GROUP BY AND HAVING
• Aggregate functions (SUM, AVG, COUNT)
• INNER JOIN operations
• Window functions such as RANKO, OVER

Conclusion

This project demonstrated the use of SQL Server for solving business problems through structured data analysis and relational querying. The analysis provided valuable insights into customer behavior, sales performance, revenue trends, music preferences, and artist productivity.

Overall, the project strengthened practical SQL querying skills, improved understanding of relational databases, and enhanced the ability to transform raw data into meaningful business intelligence suitable for decision-making.

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